

Introduction to Microeconomics
Lecture 5—Supply & Demand

Prof. Alejandro Melo-Ponce



NAZARBAYEV
UNIVERSITY

ECON101
Fall 2023

- Why do male soccer players earn more than female soccer players?
- Why do diamonds cost more than water?
- Why do Picasso's paintings sell for more than other artists?
- Is it the cost of production that determines prices, or is it willingness to pay that determines prices?
- Alfred Marshall, in *Principles of Economics*, 1890 was the first to explain clearly how both costs and willingness to pay interact to determine market prices.

Definition

The **market** for any good or service consists of all (actual or potential) buyers or sellers of that good or service.

The demand curve

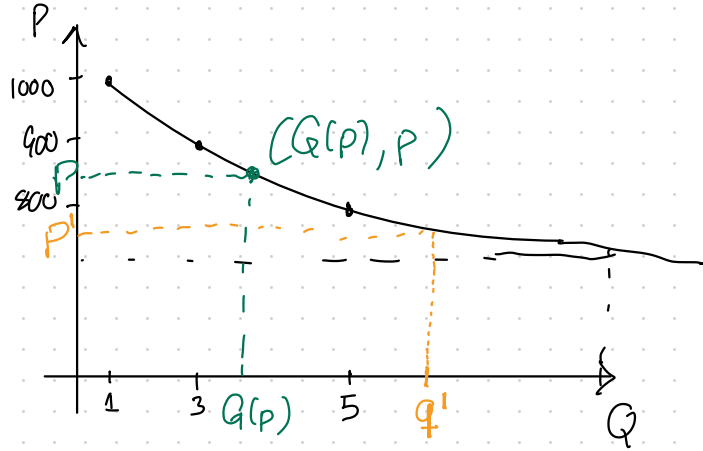
The **demand curve** for a good or service tells us the total quantity of that good or service that buyers wish to buy at each price.

Q_D (price of the good; determinants of demand)

↳ price $\mapsto$ quantity demanded Q_D

- Factors also affecting Q_D (quantity demanded).

Graphical Representation of $Q_D(P)$



Q_D = cups of coffee consumed
at NU (100) / day

Why the demand curve slopes downward?

1. As the good becomes more expensive, consumers switch to other goods.
2. As the good becomes more expensive, consumers can't afford to buy as much of it.

The demand curve is the set of all price-quantity pairs (Q, P) at which buyers are satisfied.

Horizontal Interpretation: at Price P , $Q_D(P)$ is the quantity demanded.

Vertical Interpretation: at quantity Q' , P' is the max. price that consumers could be willing to pay for the last unit $\leftrightarrow$ Marginal Benefit for the last unit.

The supply curve

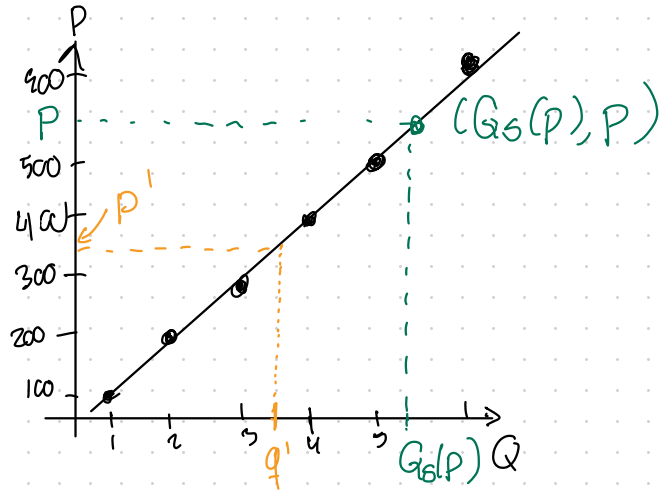
The **supply curve** of a good or service tells us the total quantity of that good or service that sellers wish to sell at each price.

$Q_s(p; \text{determinants})$

$p \mapsto Q_s$ is quantity supplied
by sellers
↓
price of
the
good.

- Determinants of supply,
– Other factors affecting
quantity supplied.

Graphical Representation of $Q_s(p)$



Q = cups of coffee sold at
NU (100) / day

Why it slopes upward?

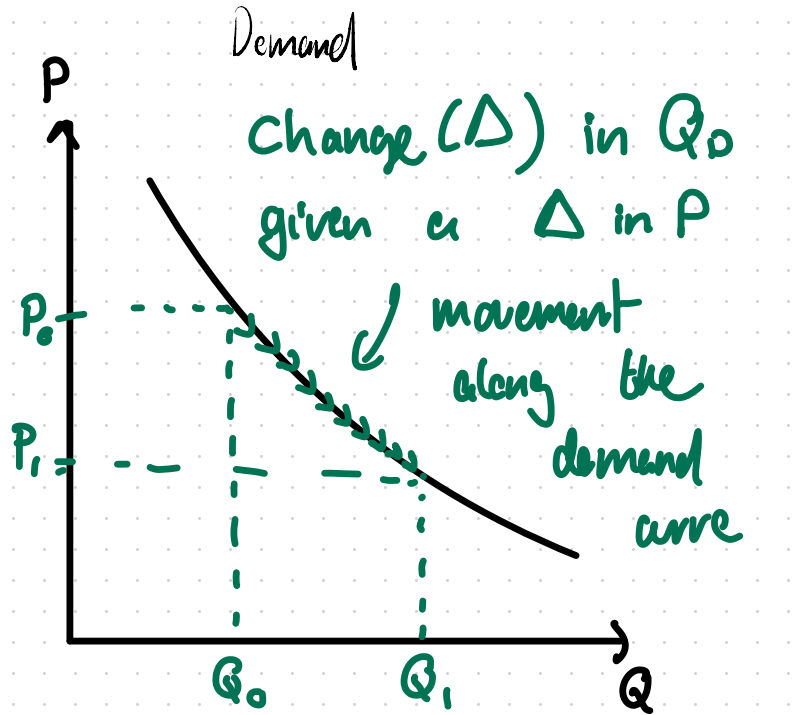
Because of increasing A.C. in production

The supply curve is the set of price-quantity pairs $(Q_s(p), P)$ at which sellers are satisfied.

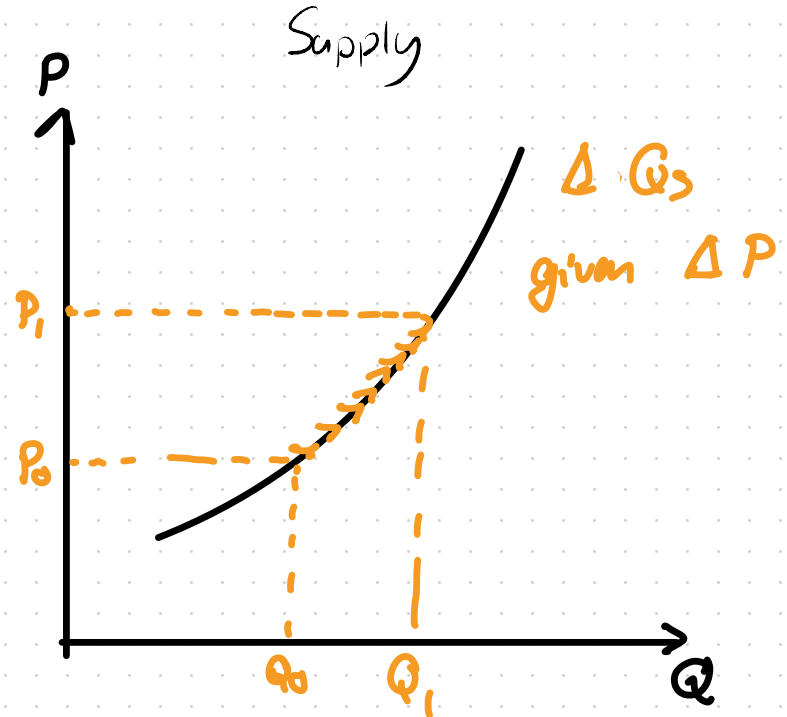
Horizontal Interpretation: at the price P , sellers are willing to sell $Q_s(p)$

Vertical Interpretation: For the last unit q' , the minimum price that the marginal seller is willing to accept is p' : Marginal Cost of that last unit.

Movements along the Supply & Demand Curves



$$Q_D = Q_D(P; \text{determinants})$$



$$Q_S = Q_S(P; \text{determinants})$$

Shifts in the Demand & Supply Curves

Demand: $Q_D = Q_D(P; \text{det})$

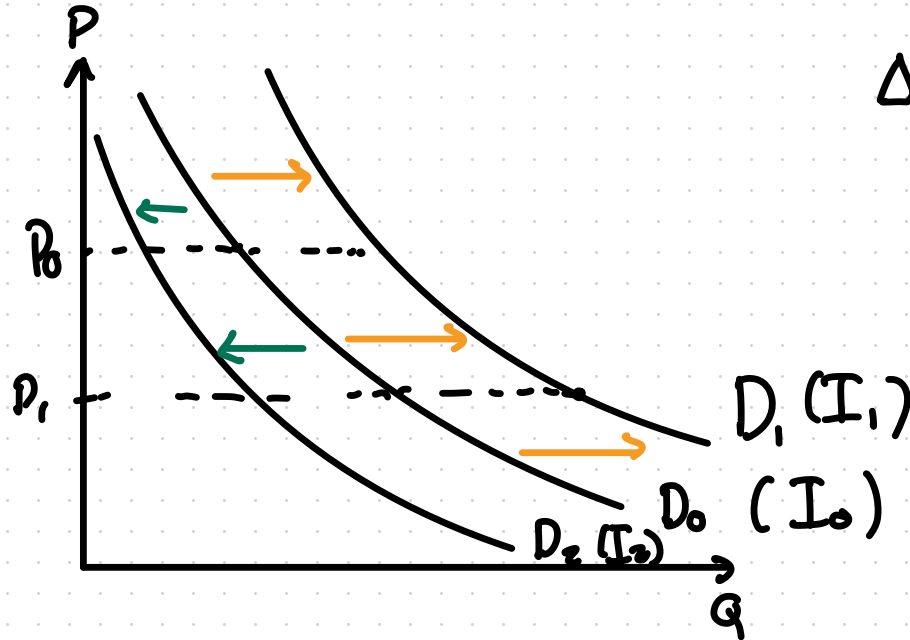
Determinants of Demand (examples)

- tastes
- prices of substitutes
- prices of complements
- Income
- Expectations of future prices
- # Consumers

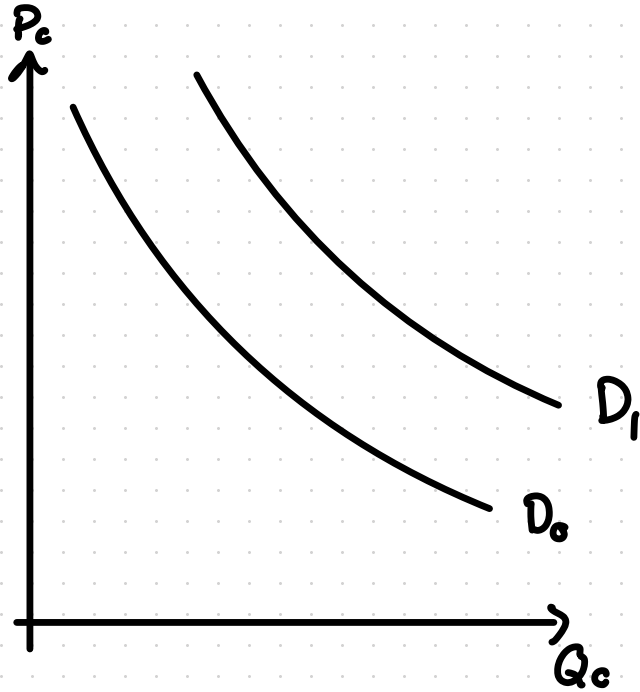
A change ΔDET in any of the determinants of demand "shifts" the demand curve.

Income (I) increases $\Delta I > 0$; $\Delta I = I_1 - I_0 > 0$

$\Delta I = I_2 - I_0 < 0$



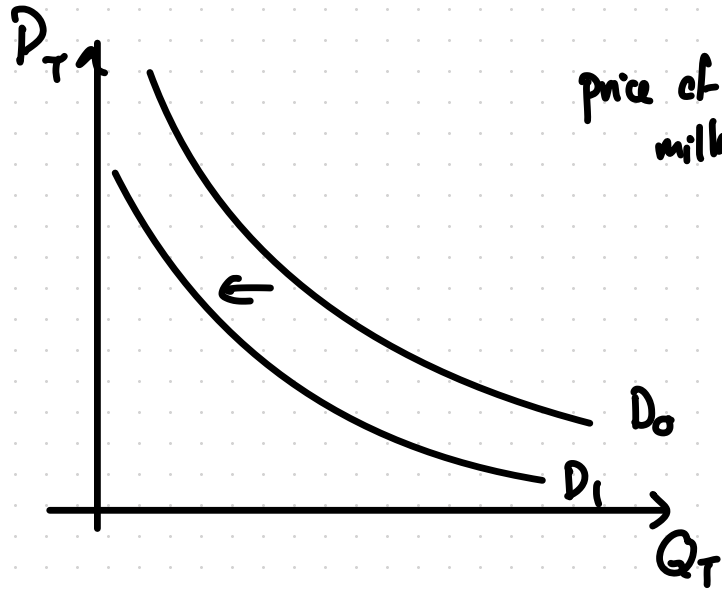
The demand for Coke.



$P_p \uparrow \rightarrow Q_{\text{Coke}}^D$ shifts
to the right

$P_p \downarrow \rightarrow Q^D \downarrow$ shifts to
the left.

Consider the demand for tea



price of
milk

$P_M \uparrow \Rightarrow Q_D \downarrow$ shifts
to the left

$P_M \downarrow \Rightarrow Q_D \uparrow$ shifts to
the right.

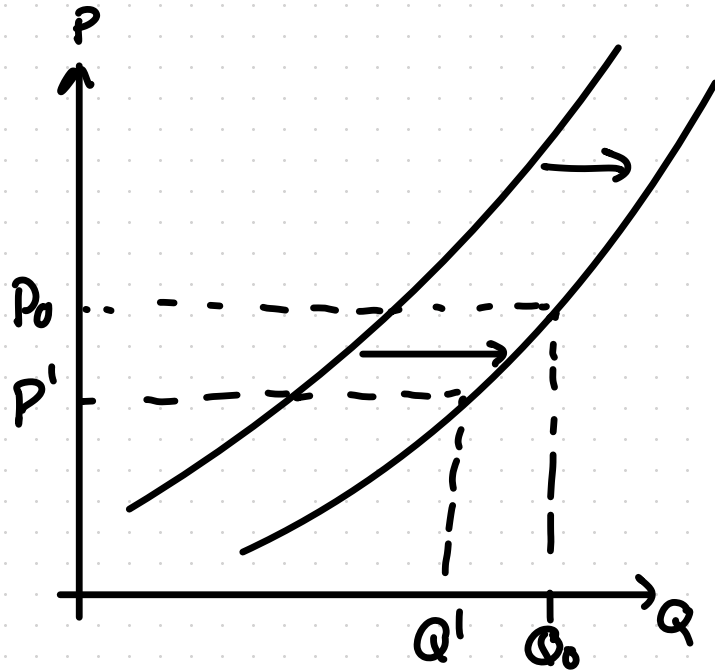
Shifts in the Supply Curve

$$Q_s = Q_s (P; \text{determinants of supply})$$

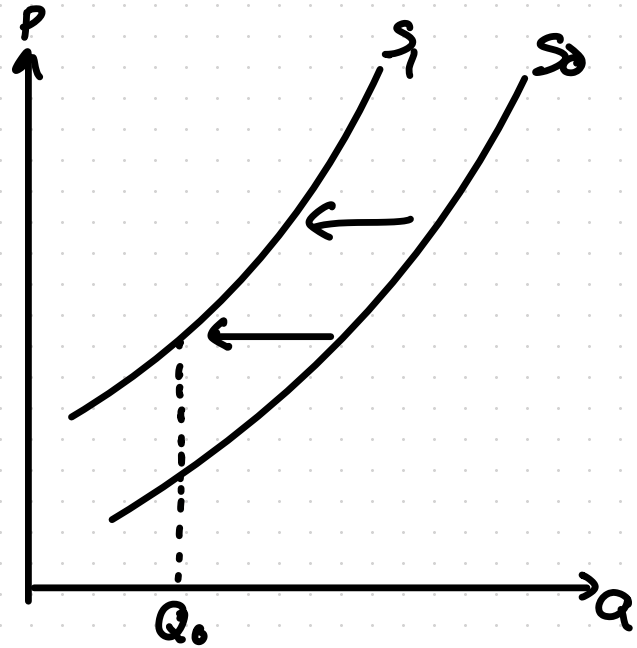
Determinants of Supply

- Costs of inputs of production
- # Suppliers
- Technology
- Expectations of future prices.
- Expectations about input prices.

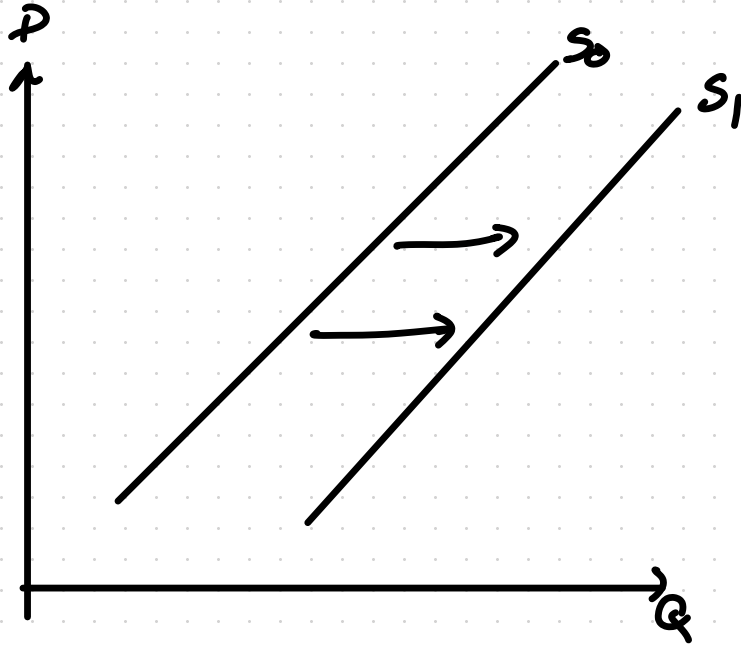
Δ Def of supply



Cost of Inputs of prod. increase



Technological Progress

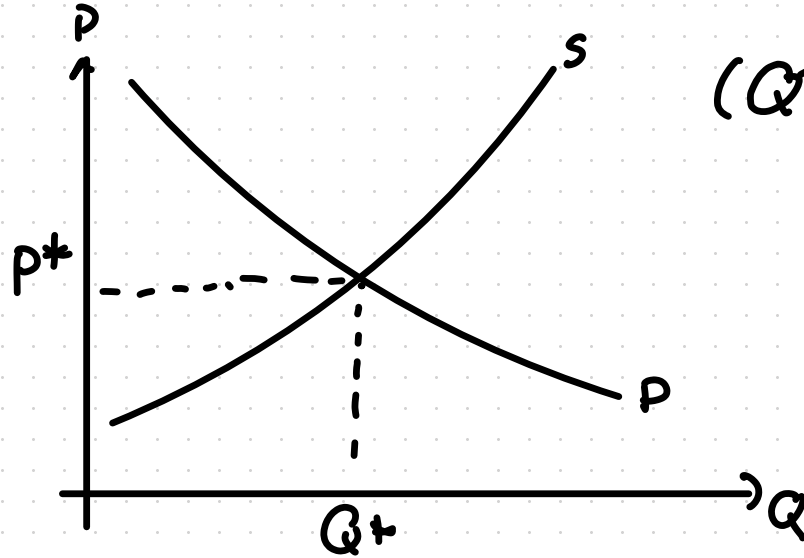


The Equilibrium Principle

A market in equilibrium leaves no unexploited opportunities for individuals, but may not exploit all gains achievable through collective action.

↳ leave it for after the Fall
break 😊

Market eq. price & quantity



(Q^*, P^*) at which

$$Q^D(P^*) = Q^S(P^*) = Q^*$$

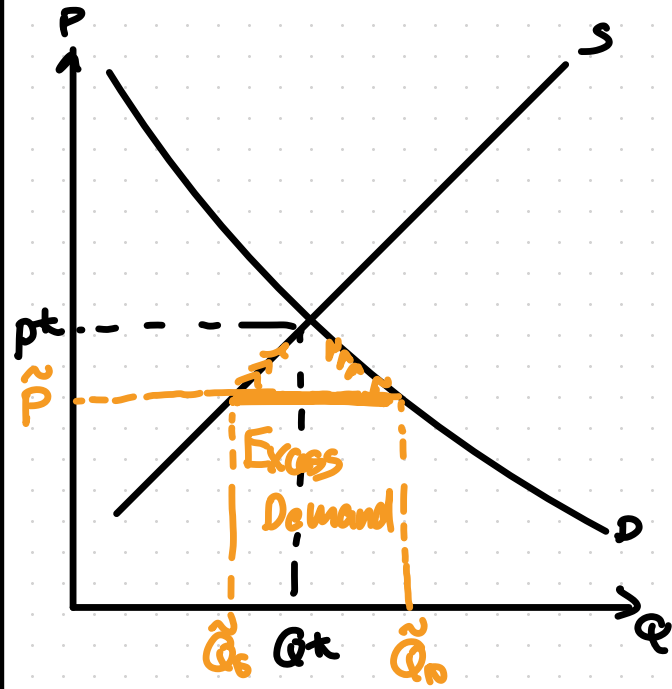
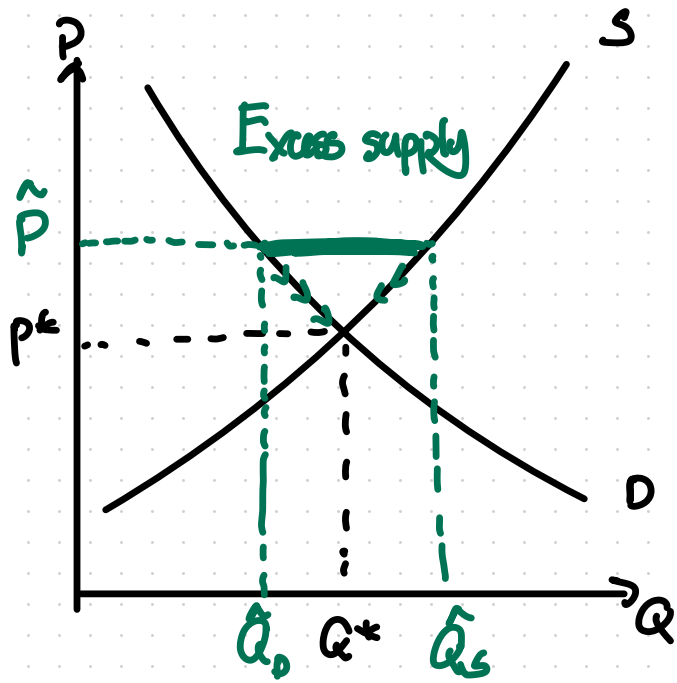
given some fixed levels
of the $DET_{S,D}$

MATH : Solve

$$Q_D = Q_D(P)$$

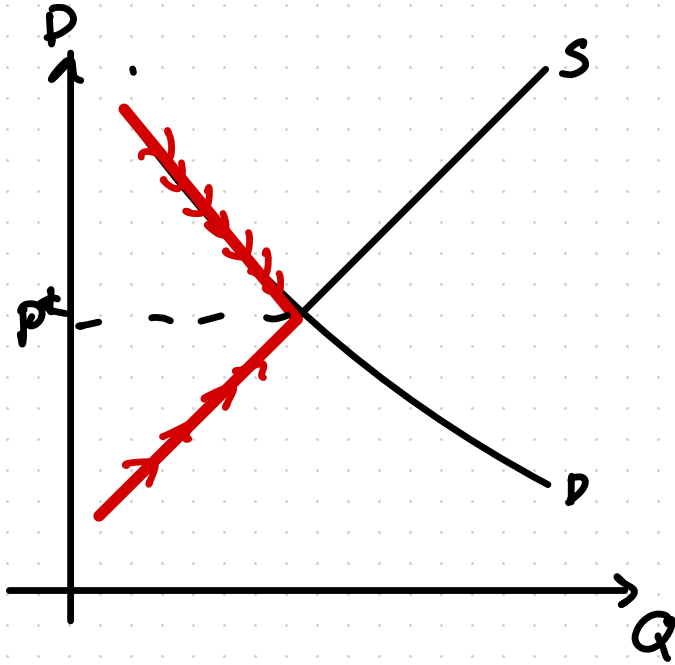
$$Q_S = Q_S(P)$$

simultaneously to find
 (P^*, Q^*)



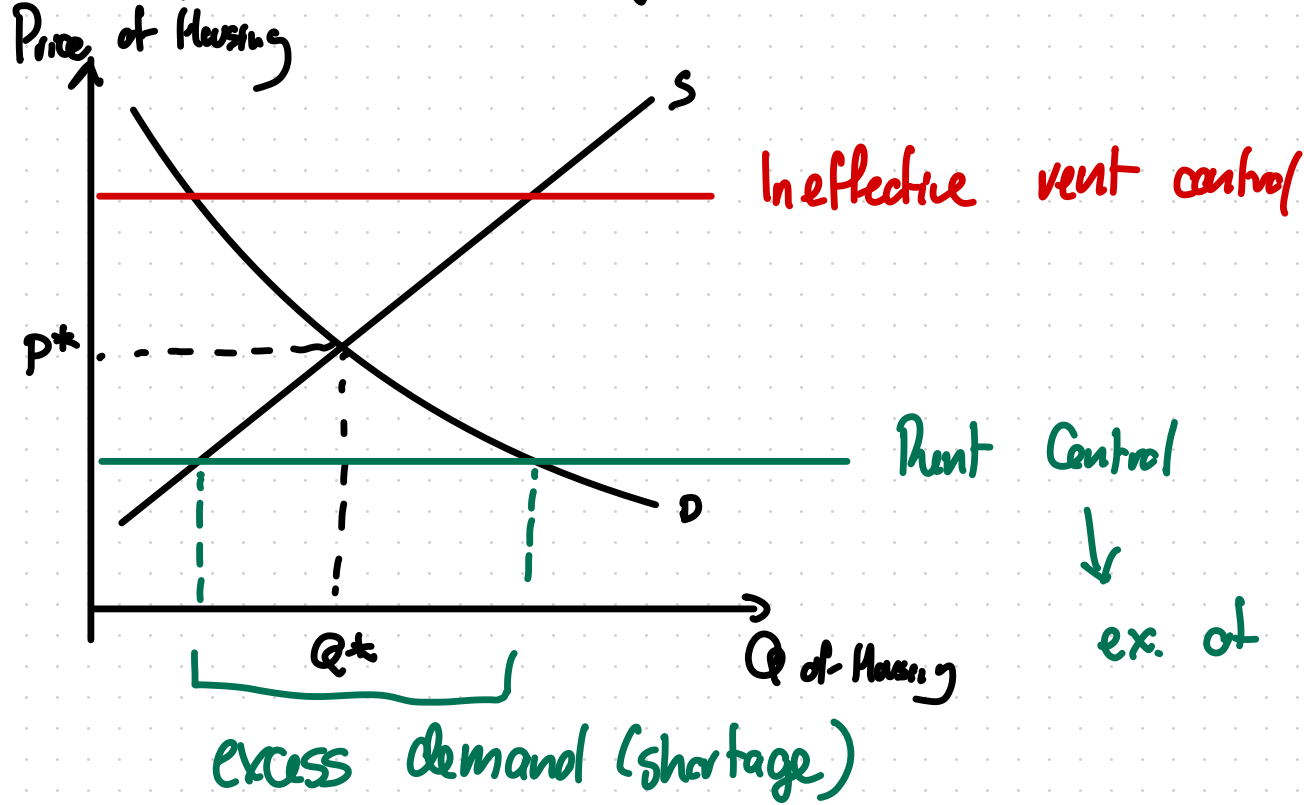
⚠ The market forces the price to change so as to reach p^*

The trading locus



— Trading locus.

What happens if we regulate prices?



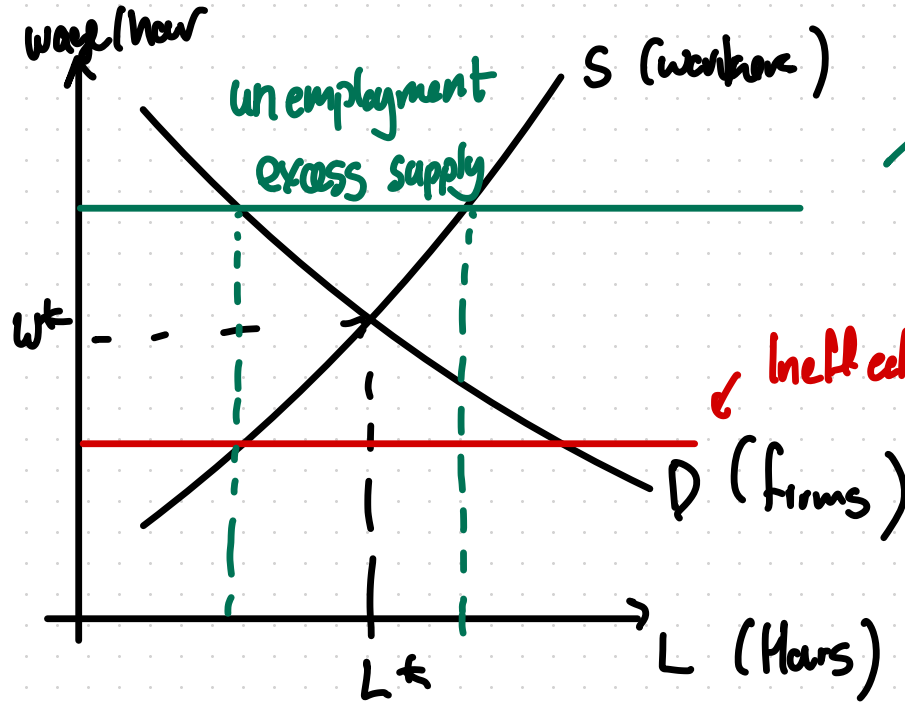
Rent Control



ex. of Price Ceiling

Minimum wage

Ex.
Price floor



Linear Demand

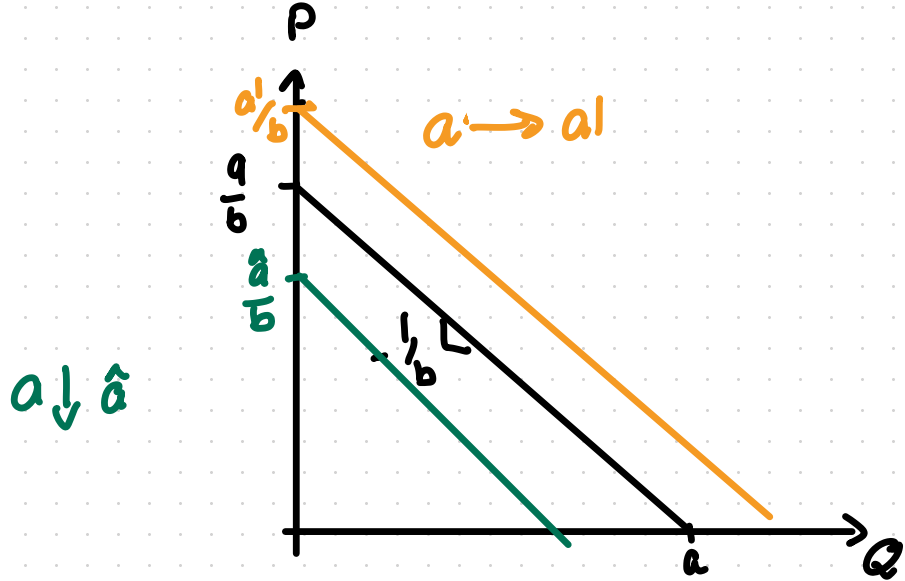
$$Q_D = a - bP$$

$$b > 0, a > 0$$

↳ Invert the demand fn.
↳ "inverse demand"

$$P = \frac{a}{b} - \frac{1}{b} Q_D$$

a is a parameter encapsulating all the demand determinants under consideration.



Linear Supply

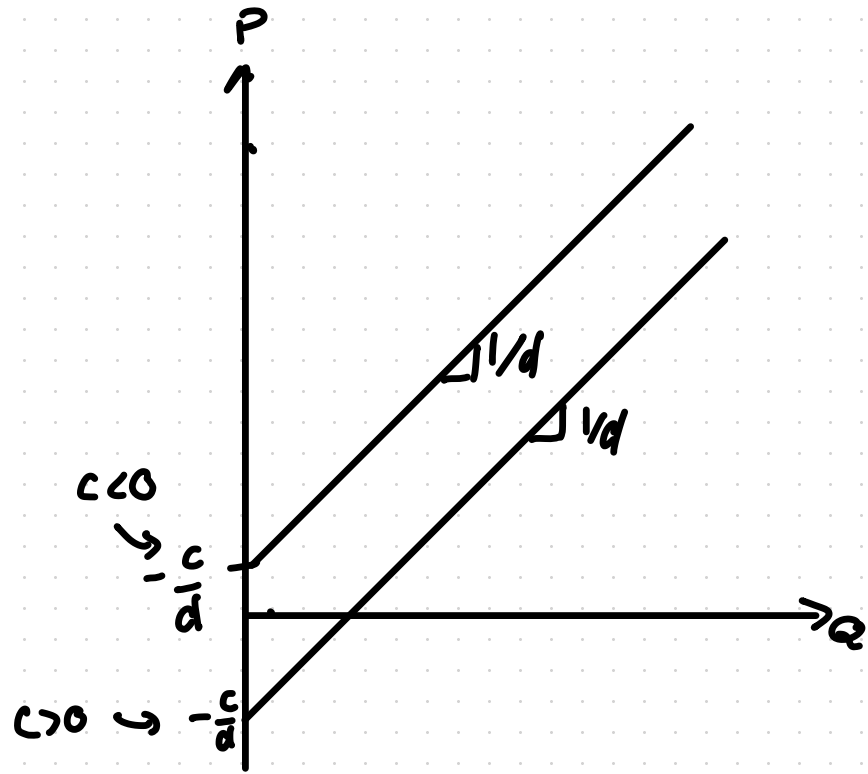
$$Q_s = C + dP$$

$d > 0$, C is a real number

To graph it, solve for $P \rightarrow$
"Inverse supply"

$$P = \frac{Q_s}{d} - \frac{C}{d}$$

* C encapsulates all
supply determinants
under consideration.



Eq. with linear demand & supply curves

$$Q_D = a - bP$$

$$Q_S = c + dP$$

For equilibrium:

$$Q_D(P^*) = Q_S(P^*)$$

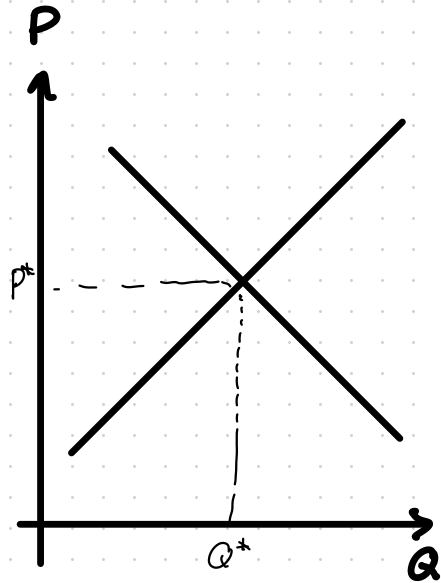
$$a - bP^* = c + dP^*$$

$$a - c = bP^* + dP^*$$

$$\frac{a-c}{b+d} = P^*$$

$$\begin{aligned} Q^* &= a - bP^* \\ &= a - b\left(\frac{a-c}{b+d}\right) \\ &= \frac{a(b+d) - ba + bc}{b+d} \\ &= \frac{\cancel{ab} + ad - \cancel{ab} + bc}{b+d} \end{aligned}$$

$$Q^* = \frac{ad + bc}{b+d}$$

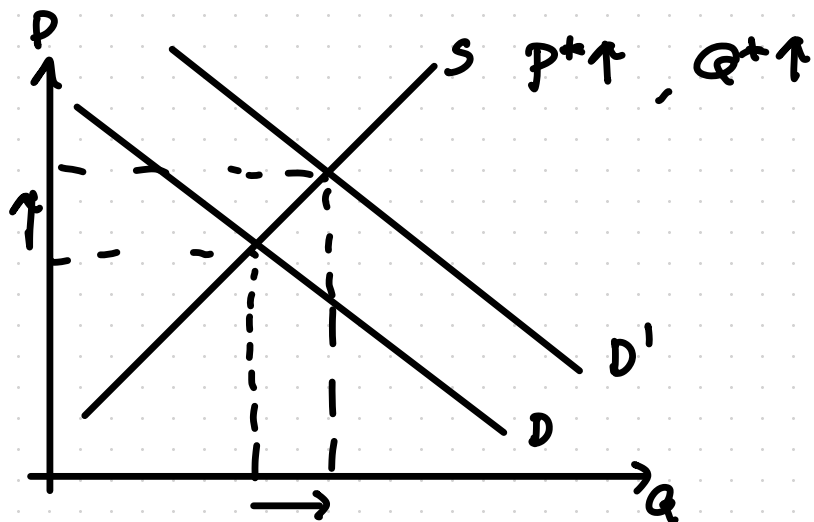


Comparative Statics (Ceteris Paribus Analysis)

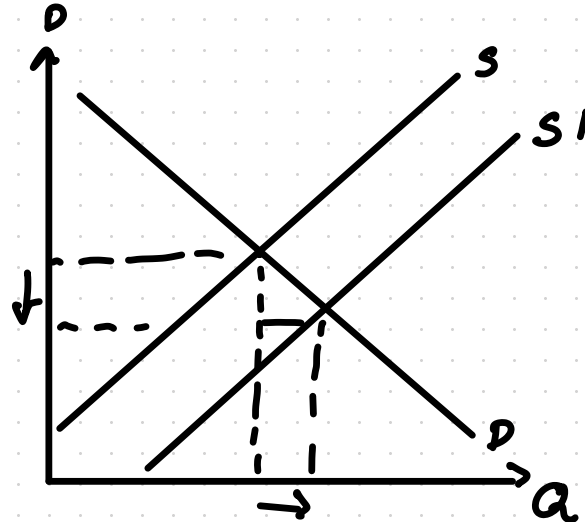
$$\frac{a-c}{b+d} = p^*$$

$$Q^* = \frac{ad + bc}{b + d}$$

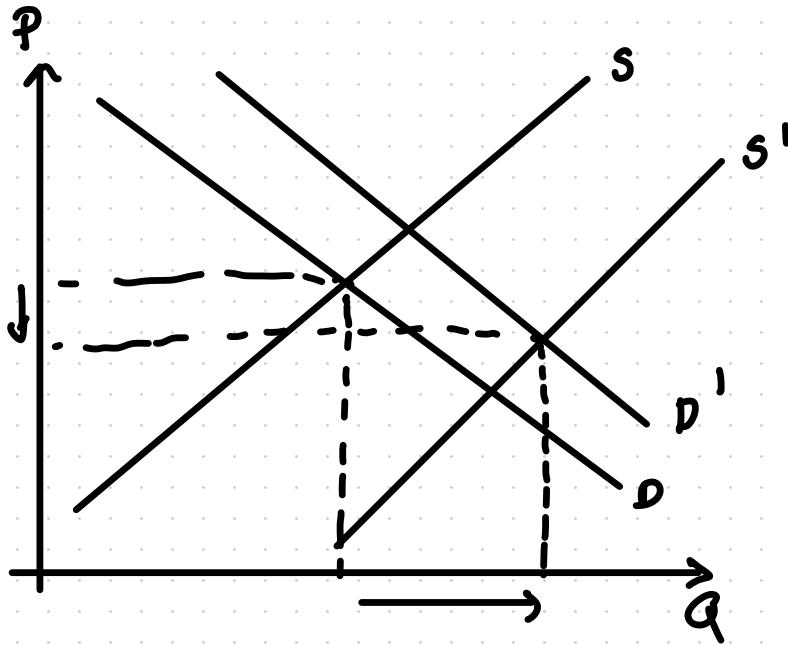
Shift of $D \uparrow \leftrightarrow a \uparrow$



Shift of $S \uparrow \leftrightarrow c \uparrow$



Changing two determinants at the same time.

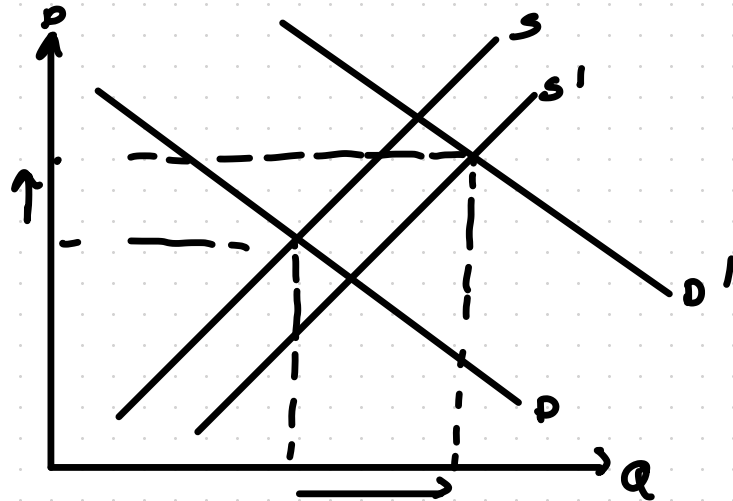


$$\frac{a-c}{b+d} = p^*$$

$$Q^* = \frac{ad + bc}{b + d}$$

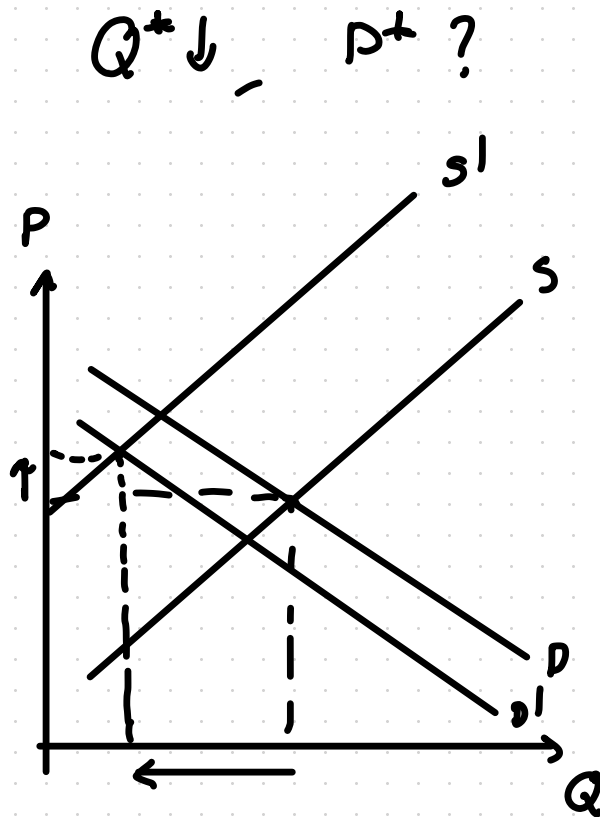
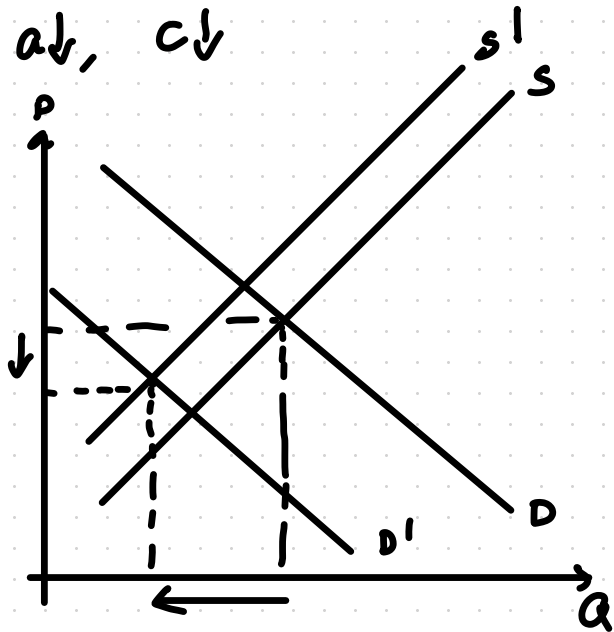
$a \uparrow, c \uparrow$

$Q^* \uparrow$, but you need to be careful about p^*



$$\frac{a-c}{b+d} = p^+$$

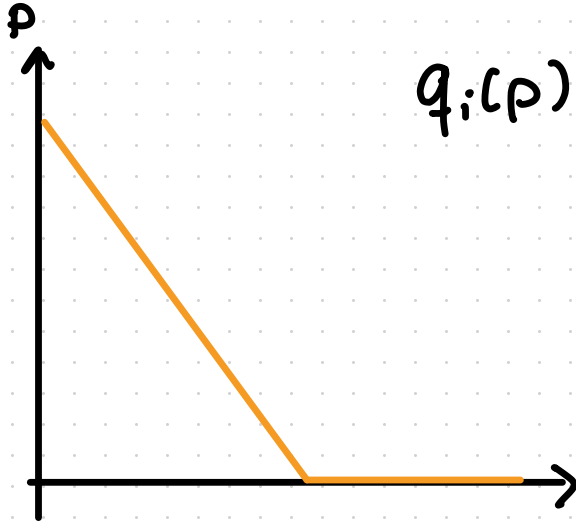
$$Q^* = \frac{ad+bc}{b+d}$$



Market Demand & Supply

Suppose individual demand curves are given

$q_i(p)$, all other determinants are held fixed.



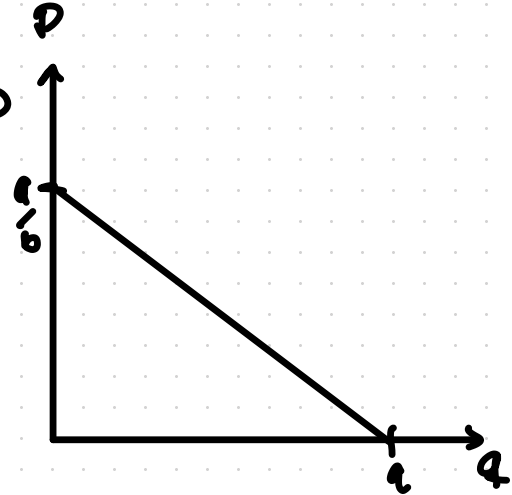
$q_i(p) \geq 0$ is the only region in which we are interested

$$q_i(p) = a - bp$$

When $q_i(p) \geq 0$?

$$\Rightarrow a - bp \geq 0$$

$$\Rightarrow \frac{a}{b} \geq p$$



Market Demand:

$$q_1(p), q_2(p), q_3(p), \dots, q_n(p)$$

Market Demand: obtained by Horizontal Summation
at price p , we add up
all demanded quantities

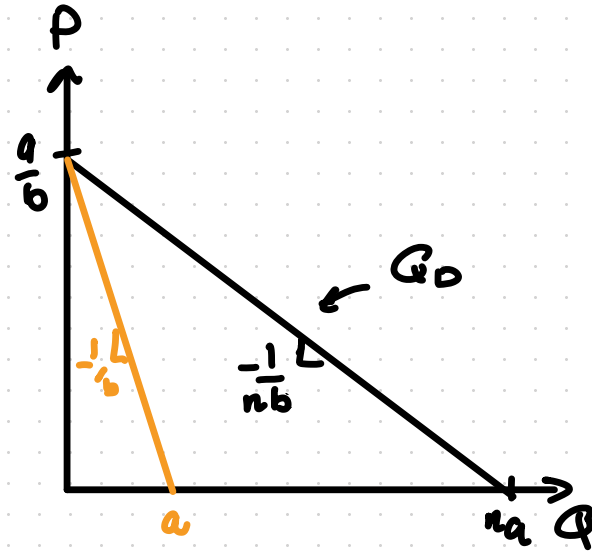
$$\begin{aligned} Q_D(p) &= q_1(p) + q_2(p) + \dots + q_n(p) \\ &= \sum_{i=1}^n q_i(p) \end{aligned}$$

Suppose we have identical demands for all i , n of them
 $q_i(p) = q(p)$ for all individuals i

$$Q_D(p) = n q(p)$$

say $q(p) = a - bp$

$$\rightarrow Q_D(p) = na - nbp$$



Example: $q_1(p) = 10 - p$, $q_2(p) = 20 - 2p$

Inverse demands

$$p = 10 - q_1 ; \quad p = 10 - \frac{q_2}{2}$$



$$\underline{Q_0(p)} = q_1(p) + q_2(p)$$

$$= 10 - p + 20 - 2p$$

$$= 30 - 3p \quad \text{for } 0 \leq p \leq 10$$

$$\hookrightarrow p = 10 - \frac{Q_0}{3}$$

Example: 1: domestic, 2: foreign, World demand

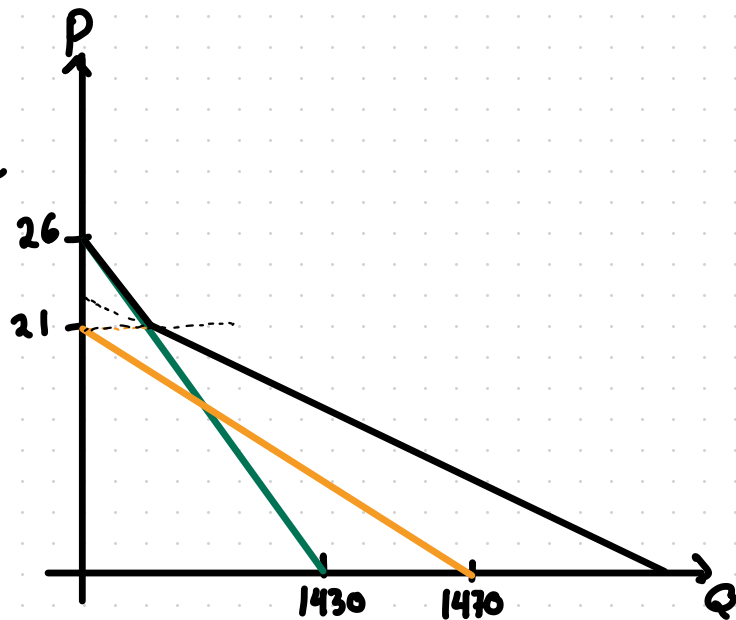
$$\underline{q_1} = 1430 - 55p; \quad p = 26 - \frac{1}{55} q_1$$

$$\underline{q_2} = 1470 - 70p; \quad p = 21 - \frac{1}{70} q_2$$

$$q_1(p) \geq 0 \Leftrightarrow 26 \geq p \geq 0$$

$$q_2(p) \geq 0 \Leftrightarrow 21 \geq p \geq 0$$

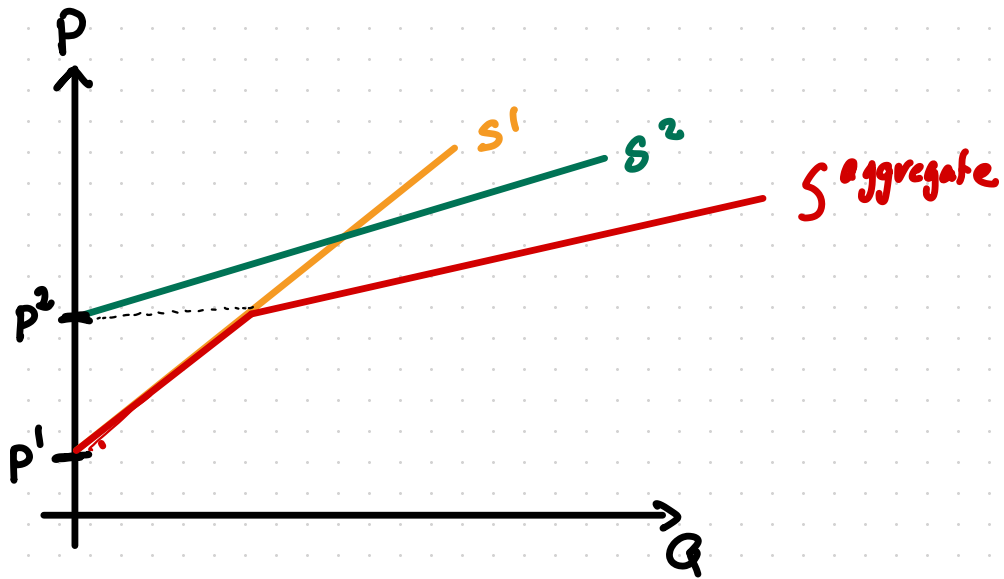
$$Q_D(p) = \begin{cases} 1430 - 55p & 21 < p \leq 26 \\ 2900 - 125p & 0 \leq p \leq 21 \end{cases}$$



Market or Aggregate Supply

$$Q_s(p) = \sum_{i=1}^n q_i^s(p) \quad \text{where } q_i^s \text{ are individual supply curves.}$$

$q_i(p)$ is only defined for $p \geq 0$ and $q \geq 0$



Example: No coffee market.

① Mrlid Consumers: $P = 900 - 90q$, $N_1 = 1000$
② Caffeine addicts: $P = 1200 - 60q$, $N_2 = 1400$

① $q_1 = 10 - P/90$

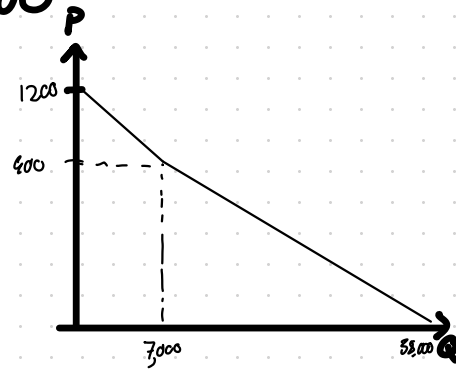
② $q_2 = 20 - P/60$

$$Q_1 = 1000 q_1 = 1000 (10 - P/90) = 10,000 - \frac{1000}{9} P$$

$$Q_2 = 1400 q_2 = 1400 (20 - P/60) = 28,000 - \frac{700}{3} P$$

$$Q_1 \geq 0 \iff 900 \geq P \geq 0$$

$$Q_2 \geq 0 \iff 1200 \geq P \geq 0$$



$$Q_0(P) = \begin{cases} 0 & \text{if } P > 1200 \\ 28,000 - \frac{700}{3} P & \text{if } 900 < P \leq 1200 \\ 10,000 - \frac{1000}{9} P & \text{if } 0 \leq P \leq 900 \end{cases}$$

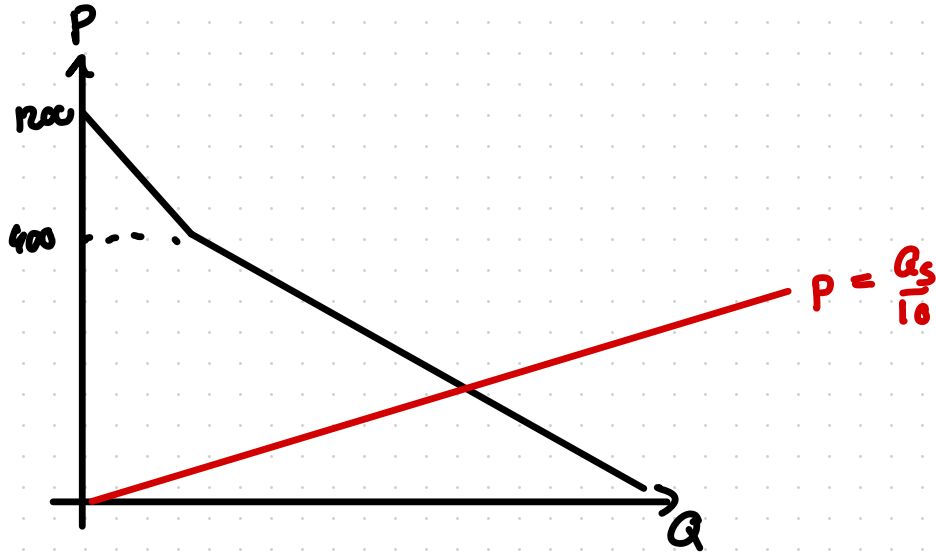
$$Q_D(p) = \begin{cases} 0 & \text{if } p > 1200 \\ 28,000 - \frac{70}{3}p & \text{if } 900 < p \leq 1200 \\ 38,000 - \frac{310}{9}p & \text{if } 0 \leq p \leq 900 \end{cases}$$

Incorrect:

$$28,000 - \frac{70}{3}p = 10p$$

↳ Homework!

Suppose $Q_S(p) = 10p$. Find the mkt. eq.



$$Q_D(p^*) = Q_S(p^*)$$

$$38,000 - \frac{310}{9}p^* = 10p^*$$

$$p^* = 855$$

$$Q^* = 8550$$

Next Steps

• Quantity

- ΔQ given ΔP for both supply & demand curves
- ΔQ given Δ Determinants, again for both

↳ Elasticity $y = f(x)$

$$e = \frac{\Delta \% Y}{\Delta \% X} = \frac{\frac{\Delta Y}{Y}}{\frac{\Delta X}{X}}$$

- unitless quantity
- measure of sensitivity